TECH BULLETIN

ELECTRODE CONSUMPTION REDUCTIONS UP TO 25% - 30%

ITC & EAFab Corporation have partnered to design, manufacture and implement a fully automated electrode spraying unit capable of reducing electrode consumption up to 30%. Funding for this project was provided through a contract by the U.S. Government, Department of Defense and The Forging Consortium for energy conservation in steel manufacturing. Over the past decade the extreme increase in electrode cost and overall supply shortages have been a common challenge for steel producers. These challenges continue to have a negative impact on operating margins and overall net profits. Current practice and the industry standard of reducing consumption, is to cool the electrode during operation by cascading water directly onto the circumference of the electrode using a water spray ring mounted directly under the clamp. While cooling the top portion of the mast only, this water addition has no effect on reducing consumption of the electrode that is submerged in the furnace. To combat these challenges, ITC has designed a coating specifically for electrodes, ITC 303EC, formulated for application on carbon/graphite substrates and can be applied at elevated temperatures.

ITC 303EC is a proprietary coating designed to work in the most extreme conditions of an EAF furnace. Using ITC 303EC, the following benefits can be achieved,

- Reduction of carbon/graphite consumption by 25% to 30%
  - Reduced side oxidation
    1) Reduced oxidation cone
    2) Increased tip diameter
  - Reduced tip consumption
    1) Reduced number of electrode connections
    2) Reduced tip loss
- Additional heats
  - Longer electrode life between additions allows for additional heats per week or month
- Improved conductivity of electrodes
  - By redistribution of the electrical current flow through the tip of the electrode
- Time savings
  - Improved working conditions for furnace operators and staff
  - Reduced crane time